



Analysis of an Uncrewed Mishap During Ground Test: A Case Study

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Abstract

On 16 November 2020, a MQ-8C Firescout was preparing for a form/fit/function ground test of a new battery. The ground test was considered low risk and therefore did not require extensive risk mitigation steps. The test points were simple: start the air vehicle, do a systems check, and shut down the air vehicle. On the morning of the test event, the MQ-8C was chained to the ground (as it was a maintenance ground turn), and a maintenance laptop was connected to interface with the air vehicle systems. A command and control communication link was not established with a ground station. The test team started the air system, and everything functioned as expected until test personnel initiated shutdown. Within 30 seconds, the air vehicle went to full power, the rotor disk shifted from parallel to the ground to full forward and full right, the shifted disk cut the only connection to the maintenance laptop, and the system ran for approximately 6 minutes until the transmission came apart causing over \$2,500,000 of damage. This paper summarizes the facts surrounding the mishap and offers conclusions and recommendations on how to avoid similar mishaps in the future.

Keywords Failure modes and effects analysis · Military aviation · Mishap causal analysis · Uncrewed aviation

1 Introduction

The United States Navy (USN) has announced that it intends to increase the number of uncrewed systems within naval aviation [1]. This has led to the establishment of various program offices (PMAs) within the Naval Air Systems Command (NAVAIR) and an uncrewed developmental test and evaluation squadron, UX-24. In July of 2019, the USN fielded the MQ-8C Firescout [2], an uncrewed helicopter similar in size to a UH-1. Figure 1 is an image of a MQ-8C Firescout operating from the flight deck of USS Montgomery (LCS 8). PMA-266 is the program office assigned to the acquisition process for the Firescout. As part of the life cycle management of the system a decision was made to upgrade the lead acid battery to lithium-ion (Li-ion) battery. UX-24 was assigned the task of developmental test and evaluation of the new battery.

On 16 November 2020 UX-24 conducted a form/fit/function ground test on one of their assigned MQ-8Cs. The

ground test was considered low risk and therefore did not require extensive risk mitigation steps. In accordance with the approved test plan, the test aircraft was chained to the ground to prevent inadvertent flight. A Maintenance Portable Electronic Display Device (MPEDD), a maintenance laptop, was used to interface and control the air vehicle during the test event. A command and control communication link was not established with a ground station. All test points proceeded as planned until the test team initiated shutdown. Within 30 seconds of initiating shut down the air vehicle went to full power, the rotor disk shifted from parallel to the ground to full forward and full right and cut the connection to the MPEDD. The system continued to run for approximately 6 minutes until the transmission came apart causing over \$2,500,000 million dollars of damage.

What caused this multi-million dollar loss during what was defined as a low-risk ground test? The information presented in this paper was derived from the command investigation, commonly referred to as the Judge Advocate General Manual (JAGMAN) investigation [4]. The JAGMAN investigation was conducted in parallel to the formal Class A aviation mishap investigation. The opinions, recommendations, and lessons learned presented in this paper are those of the author and do not represent the official position of the NAVAIR or the USN.

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Fig. 1 MQ-8 operating from the flight deck of USS Montgomery (LCS 8) [3]

This paper analyzes the facts surrounding the mishap and identifies lessons learned to prevent similar mishaps in the future. The intent of publishing these details in an academic setting is to help initiate a conversation relating to uncrewed systems development and risk mitigation planning. Complicated systems fail in complicated ways. By removing the human from the loop, system designers have removed a critical risk mitigation element from developmental flight test. When events transpire that were not explicitly planned for, software algorithms may not be robust enough to prevent a catastrophic mishap. This is especially true in developmental test and evaluation.

This paper is structured as follows. Section 2 contains an overview of the USN mishap investigations and airworthiness processes. Additionally, it offers a short literature review. Section 3 contains the facts surrounding the mishap. Section 4 details opinions, recommendations, and lessons learned from the mishap. Section 5 summarizes the paper and offers ideas for follow on research.

2 Background

This section is structured as follows. Section 2.1 details the USN mishap investigation process. Section 2.2 provides a short overview of the USN's airworthiness certification process. Section 2.3 contains a short literature review on mishap investigation and issues related to Li-ion battery hazards.

2.1 USN Mishap Investigation

The formal USN mishap investigation process is dedicated to preventing future mishaps [5]. The investigations are geared towards identifying the root causes of the mishap in an attempt to ensure that they will not be repeated. To

accomplish this, the investigators are given a unique tool. They are authorized to provide privilege during their investigation. Privilege entails collecting data that is considered non-attribution in nature. This means that anything a witness provides to investigators during a privileged statement cannot be used against the individual for disciplinary actions. The safety investigation is led by a senior naval officer (O-5 or above) who is a graduate of the USN Aviation Safety Officer (ASO) School and the team is given access to extensive resources to help them accomplish their mission.

USN mishaps are separated into multiple categories. The most severe is identified as a Class A mishap. Class A mishaps involve the loss of life, loss of an aircraft, or over \$2,500,000 million dollars in damages [5]. As previously stated, the USN mishap process is designed to identify causal factors in an attempt to prevent future mishaps. For Class A mishaps the USN normally performs a parallel investigation, a JAGMAN [6]. The purpose of the JAGMAN is to document the facts behind the mishap and form conclusions and recommendations separate from the formal, privileged, aviation safety investigation. The JAGMAN is also led by a senior naval officer (O-5 or above). The only data permitted to be shared by the two investigations is factual-based data (e.g., aircraft maintenance records, and air data recordings).

The data used for this paper was from the JAGMAN performed on the 16 November 2020 Class A mishap [4]. The investigating officer was a senior naval officer, a graduate of the ASO school, a graduate of The United States Naval Test Pilot School (USNTPS), and had performed extensive research into airworthiness certification of uncrewed systems within the USN [7–9].

2.2 Navy Airworthiness Process

The airworthiness authority for naval aviation resides with the Commander of NAVAIR, a three-star admiral. NAVAIR delegates this authority to the Airworthiness and CyberSafe Office (ACO). The process used by the ACO to grant flight clearances is outlined in NAVAIRINST 13034.1G, Airworthiness Policy [10]. When evaluating a system for a flight clearance the ACO determines the environment the system will be used in. They will then identify several technical area experts (TAEs) that would be required to sign off on a flight clearance. The TAEs then gather a number of standards to evaluate the system against and the methods of compliance for those standards. Following test and evaluation (i.e., software in the loop, laboratory, ground, and flight) the TAEs identify areas of risk. Risk mitigation steps are then taken, or not, depending on the hazard identified. A flight clearance would then be issued at the appropriate level based on the amount of perceived risk.

There are two types of flight clearances, Permanent Flight Clearance (PFC) and Interim Flight Clearance (IFC). The purpose of an PFC is to grant an envelope for qualified fleet aircrew to safely operate an air vehicle with limited safety mitigation steps in place. Typically the envelope is restrictive to allow the aircrew some safety margin before an exceedance would have catastrophic consequences. One example of this may be the maximum airspeed limit. A hypothetical clearance envelope may be to operate the air vehicle under 300 knots at sea level even though the air vehicle could operate up to 350 knots without any issues. The purpose of an IFC is to allow a demonstration or test program to evaluate a new system while defining where the actual boundaries are. Before a developmental test (like the ground test performed by UX-24 on 16 November 2020) the ACO will grant an IFC for the configuration being evaluated. The IFC will normally identify several risk mitigation steps (i.e., real-time data monitoring or a restrictive operating envelope) that will be required during the test event.

2.3 Literature Review

When attempting to find the root cause of a mishap, ASO graduates are taught to analyze a mishap via Reasons' model. Reasons' model is commonly referred to as the Swiss cheese model [11–15]. The model has become the standard by which all aviation mishaps are investigated by the USN. It is a simple model that can be applied to multiple situations. It can be explained by an analogy, that of multiple slices of Swiss cheese stacked together. Each layer of Swiss cheese represents a layer of risk mitigation steps that may be taken prior to the mishap. Each layer has multiple holes, just like a slice of Swiss cheese. The holes represent unidentified risk (the holes may be large or small), or residual risk (typically smaller holes as there were most likely risk mitigation steps taken), that exist. It is only when the holes line up that a mishap is

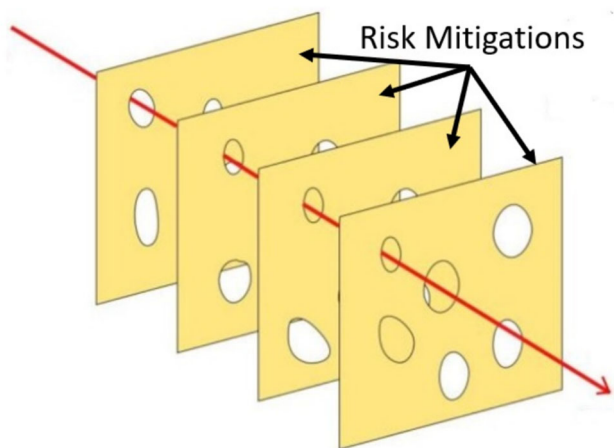


Fig. 2 Graphical Representation of Reason's Model, Adapted from [12]

possible. Figure 2 is a graphical representation of Reason's model.

On the day of the mishap, UX-24 was performing a developmental ground test on a replacement battery. In 2020 the fielded MQ-8C was equipped with a lead acid battery. As part of an upgrade plan, PMA-266 began a program to acquire and field a Li-ion replacement battery. The batteries seemed like an ideal replacement. However, PMA-266 identified a risk relating to Li-ion batteries exploding when they had a malfunction [16–20]. As part of the statement of work for the replacement battery program, PMA-266 required risk mitigation steps for the Li-ion battery as part of the upgrade program [4].

3 Facts Surrounding the Mishap

This section is structured as follows. Section 3.1 details the events leading up to the day of the mishap. Section 3.2 details the events that transpired on the day of the mishap. All factual data contained in this section came from the JAGMAN investigation [4] unless otherwise noted.

3.1 Events Leading up to 16 November 2020

The MQ-8C reached initial operating capability (IOC) in 2019 [2]. The Firescout IOCed with a lead acid battery. This was considered a simple way to provide the necessary power for the aircraft systems during operation. However, these batteries were sub-optimal. In an effort to upgrade the MQ-8C, PMA-266 began a program to acquire, test, and field a Li-ion replacement battery. As part of the procurement process, PMA-266 levied a requirement on the prime contractor for the new Li-ion battery in an attempt to mitigate the explosive risk inherent in Li-ion batteries. PMA-266 required the new battery to have a safeguard to prevent it from drawing an excessive amperage to mitigate the risk. The prime contractor elected to install software that would briefly shut down the battery if it detected an over-amperage condition (over 20 amps). PMA-266 accepted this risk mitigation step and acted as the lead system integrator for the Li-ion battery upgrade.

Before the planned developmental test and evaluation on the UX-24 MQ-8C, PMA-266 ensured the replacement battery underwent qualification testing. All of the failure modes of the battery were analyzed and the battery passed all of its tests. It was then delivered to UX-24 for the test program.

As part of any developmental test and evaluation program, the ACO identifies the proper TAEs to evaluate a development test plan for its IFC. The 16 November 2020 UX-24 test plan was no exception. Before any developmental test event, UX-24 leadership evaluates the event to see what level of risk exists. After reviewing the ground test plan for the Li-

ion battery, and the IFC, UX-24 identified the event as low risk. The ground test involved a fit check, system power-up, pre-flight checks, engine oil flow rate checks, engine start, and engine shutdown.

For maintenance ground turns (engagement of the rotors and run up to ground RPM) of the MQ-8C it is an approved standard operating procedure to chain the vehicle to the ground to prevent inadvertent flight. This enables maintenance personnel to execute the run-up without the presence of a Naval Air Training and Operating Procedure Standardization (NATOPS) qualified Air Vehicle Operator (AVO). On the morning of 16 November 2020, the mishap vehicle was towed to the turn spot and prepared for the test event. As there was no intent for flight the mishap vehicle was chained to the ground. Flight test personnel from UX-24 then prepared the mishap vehicle for the first Li-ion ground turn of the MQ-8C. All maintenance actions on the mishap aircraft were performed per established maintenance procedure, and the mishap vehicle was considered ready for test. No major deficiencies were noted that would prevent the ground turn. The 16 November planned test event would be the first time the test Li-ion battery would be used for a full maintenance start-up and shutdown of the MQ-8C [4]

3.2 Start up Through Shutdown on 16 November of the Mishap Vehicle

On the morning of 16 November 2020, a MPEDD was hardwired to the test aircraft. As it was a maintenance

turn, a command-and-control communication link was not established, or planned, with a remote Mission Control Station because there was no requirement for a NATOPS-qualified AVO during the MPEDD-controlled ground turn. The MPEDD was connected to the ground control panel (GCP) on the mishap vehicle through a wire. Ground test maintenance personnel started the mishap vehicle using the MPEDD at 09:55:19 (all times are reported in local, Zulu + 4) in accordance with established procedures. At approximately 10:08:43 all ground test points were complete and ground test maintenance personnel commanded shutdown via the MPEDD.

During a normal shutdown, the MQ-8C logic between the Vehicle Management Computers (VMCs) and Power Lever Actuators (PLAs) is designed in such a way that numerous checks are completed to ensure a safe shutdown. However, the first shutdown of the MQ-8C with the replacement battery did not follow the normal sequence.

At 10:08:43.358 the PLAs were commanded to cut off. Shortly thereafter the generators were disabled. When the generator went offline, the battery transitioned from a charge state to a discharge state by design. In a discharge state, the battery is exposed to the generator bus. At the time of shutdown, the generator bus load was 25 amps. The current regulation software inherent to the test Li-ion battery sensed the perceived over-amperage condition and took the battery offline for approximately 900 ms.

At 10:08:43.627 the zero voltage condition resulted in a complete power loss to the VMC and flight control actuators

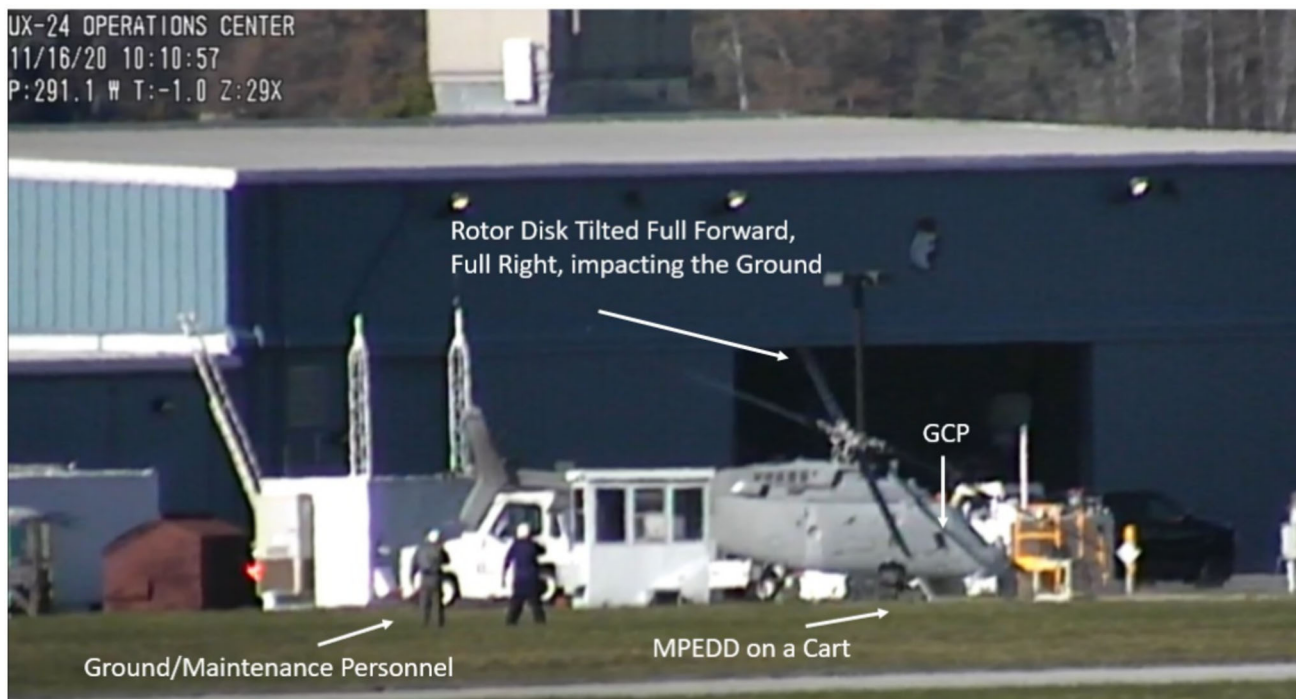


Fig. 3 Image of the Mishap Vehicle taken at 10:10:57 [4] with multiple items identified

holding the rotor disk parallel to the ground. It normally takes 0.6 seconds for the PLA to acknowledge receipt of a shut-off command. the PLA was not able to acknowledge receipt based off the power loss to the VMC that occurred. This put the test vehicle into an unknown and unanticipated situation that is not covered in MQ-8C NATOPS manual (aircraft manual). The engine was operating with the VMC disabled during a normal shutdown while chained to the deck.

The PLAs did not fully execute the shutdown command and were still operating when the VMC lost power. The author of the JAGMAN believed that when the VMCs went off line the electronic engine controls (EECs) also went offline which would revert the air vehicle into backup engine control (BEC). During BEC, the engine is commanded to 100% (by design). Additionally, he felt that when power was lost to the electro-mechanical actuators the auto-backlash springs were the only force applied to the actuators. The springs forced the spinning rotor full forward and full star-board once power was lost to the actuators. This resulted in the tilting of the rotor disk approximately 45 degrees down. In this position, the rotor disk blocks access to the GCP and will make contact with the ground

At approximately 10:08:50, after hearing the mishap vehicle go to high power after commanding a normal shutdown, ground test personnel attempted an emergency shutdown command via the MPEDD. At this time, they noted that the rotor tip path was making contact with the ground, had cut the wire connecting the MPEDD to the GCP, and the cart that was holding the MPEDD began rolling toward the mishap vehicle. Ground test personnel then backed away from the mishap air vehicle and attempted to determine a course of action. Figure 3 is an image depicting the mishap air vehicle at 10:10:57 on the day of the mishap with key elements identified in the image.

For approximately six minutes the rotor turned at 100% RPM with rotor tips contacting the ground and applying loads to the rotor hub for which the system was not designed. At 10:14:30 the gear system began to fail, and a substantial tail boom excitation began. At 10:14:35 the rotor head began to come apart. At 10:14:38 the tail boom broke off the mishap vehicle. At 10:14:41 the transmission appeared to fail and the main rotor blades were thrown. The engine continued to operate until ground test personnel initiated shutdown via the GCP at 10:16:10. Figure 4 is an image captured as the transmission appeared to fail. Figure 5 is an image depicting a ground test personnel initiated shutdown via the GCP.

4 Opinions, Recommendations, and Lessons Learned

This section is structured as follows. Section 4.1 contains the opinions and recommendations outlined in the JAGMAN



Fig. 4 Image of the Mishap Vehicle as the Transmission Failed, taken at 10:14:41 [4]

based on the facts that were outlined in Section 3.2. Section 4.2 contains lessons learned from the mishap. All factual data contained in this section came from the JAGMAN investigation [4] unless otherwise noted.

4.1 Opinions and Recommendations

All actions taken by the squadron, test team, and the ACO followed established procedures and processes. All actions taken by the maintenance and ground personnel in relation to the mishap followed established procedures and human factors did not play a role in the mishap.

The primary causal factor that precipitated the mishap was an insufficient level of understanding of the inherent behavior and protective measures of the test Li-ion battery and how it's performance might differ from the lead acid battery normally installed in the mishap vehicle by the lead system integrator. The qualification testing of the new battery was not robust enough to completely duplicate the air vehicle during the shutdown sequence. In particular, the battery switching from a charge state to a discharge state with an excess of 20 amps on the generator bus.

The possibility of a loss of power during the air vehicle shutdown sequence between the shutdown command and the execution of the PLA off command, approximately 0.6 seconds, was not accounted for by the systems integrator.



Fig. 5 Image of ground test personnel shutting down the air vehicle via the GCP, taken at 10:16:09 [4]

When the VMC (both) went off-line the EEC (both) also went offline reverting the mishap vehicle most likely went into BEC mode. When power was lost to the electro-mechanical actuators the anti-backlash springs were the only force being applied to the actuators. The springs forced the spinning rotor full forward and full starboard once power was lost to the actuators. This blocked access to the GCP. As the wire connecting the MPEDD to the GCP was severed shortly after commanding shutdown, there was no way to communicate with the mishap vehicle in a timely manner to avert the mishap.

The MQ-8C NATOPS manual states: In the event of a true dual VMC failure, or failure in which both VMCs become unstable, the air vehicle will not be controlled and will crash. In this case, flight termination is the best option. The events of 16 November 2020 put the mishap vehicle in a Dual VMC failure state while chained to the ground with no way to communicate with the air vehicle while it was turning at 100% RPM.

The author of the JAGMAN recommended that prior to continued developmental test with the Li-ion battery, a fresh failure modes and effects analysis should be performed. The one performed as part of the acquisition process was lacking.

This is a perfect example of Reasons' Model for aviation mishap analysis [11–15]. If there had been a risk mitigation step in place for any of the aspects in the chain of events the mishap could have been averted.

4.2 Lessons Learned

As we attempt to expand the use of uncrewed vehicles (across all domains) there are numerous lessons that can be learned from the events of the 16 November 2020 Firescout mishap.

- Complicated systems fail in complicated ways: This lesson has been learned multiple times during flight test within naval aviation. Modern aircraft have multiple systems working together to accomplish their mission. This can make the system interactions extremely complicated. When things start to fail, they tend to fail in complicated ways.
- Uncrewed doesn't make it easier: One of the requirements of USN development test is that a trained engineering test pilot is required to be in control of the air vehicle during test flights. An engineering test pilot (a graduate of an internationally recognized test pilot school) has been trained on how to test experimental aircraft. They generally make the right decision when the unexpected happens during flight test. If we are going to remove the human from the loop in uncrewed aircraft we will be removing an inherent risk mitigation element in developmental test and evaluation. In the case of the MQ-8C mishap, there was not a test pilot in the vehicle to ex-

cute a manual shutdown, nor was there a test AVO in a position to make inputs to the mishap vehicle.

- Test aircrew need to be ready for their new role: Uncrewed flight test is still relatively new. Most flight test teams are still led by a graduate of test pilot school. Most of them have little (if any) experience in uncrewed aviation. It is important for the test pilot community to understand their new role in uncrewed flight test. They may be the only member of the team who has been trained on developmental test and evaluation. They need to be ready to ask the hard questions and cancel the test event if they do not like the answers they are given. For manned flight test, the pilot has this no-vote capability. As testers are assigned to uncrewed test teams, they need to bring the same level of scrutiny to their new role.
- Hindsight is 20/20: No one identified the risk. No one thought it could happen. No one would have approved the test plan as a CAT-A low-risk ground test if there was an identified risk that this could happen. The simple answer is to throw resources at the test planning and acquisition process. However, there are only so many resources to go around. PMA-266 is relatively small compared to PMA-265 (F/A-18 program office) or PMA-290 (P-8 program office), and their respective budgets are roughly proportional to their size. Had there been infinite resources for this test program they may have identified the risk. As we field new systems leadership needs to understand that there are risks that may not be accounted for.

5 Conclusion

There is a famous adage in United States naval aviation: NATOPS manuals are written in blood. This refers to the fact that after every mishap the lessons learned are ultimately used to make changes to the aircraft manuals. These changes normally come in the form of Notes, Warnings, and Cautions. When students study their aircraft they take special notes of these items.

This paper highlighted an important lesson that needs to be learned by the developmental test and evaluation community. By removing the human from a position to make inputs to the air vehicle when off-nominal conditions occur we have removed a vital risk mitigation tool. The mishap analysis showed that a risk mitigation step relating to the replacement battery ultimately created a situation that was not planned for. Had there been a human in a position to make an input to the air vehicle then the mishap may have been avoided. The USN was lucky that no one was injured when the MQ-8C came apart.

Future research that analyzes uncrewed aircraft for issues relating to not having a human in or on the loop will be

required to help reduce the risk during developmental test and evaluation.

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References

1. Tucker, P.: Drones Could One Day Make Up 40 Percent of a Carrier Air Wing, Navy Says (2021). <https://www.defenseone.com/technology/2021/03/drones-could-one-day-make-40-carrier-air-wing-navy-says/172799/>
2. Reim, G.: US Navy Declares IOC for MQ-8C Fire Scout Unmanned Helicopter (2019). <https://www.flightglobal.com/helicopters/us-navy-declares-ioc-for-mq-8c-fire-scout-unmanned-helicopter/133473.article>
3. Hardgrove, S.: USS Montgomery (LCS 8) Conducts MQ-8C Operations (Image 15 of 19). Defense Visual Information Distribution Service. <https://www.dvidshub.net/image/7164454/uss-montgomery-lcs-8-conducts-mq-8c-operations> (2022)
4. Allen, R.: Freedom of Information Act Case DON-NAVY-2023-005351 (2024)
5. OPNAV M.5102.1 MCO 5100.29C VOL 9, Navy and Marine Corps Safety Investigation and Reporting Manual. "Department of the Navy Issuances" (2021)
6. JAGINST 5800.7G, Manual of the Judge Advocate General (JAGMAN). "Department of the Navy Issuances" (2021)
7. Costello, D.H., Xu, H.: Generating certification evidence for autonomous aerial vehicles decision-making. *J. Aerosp. Inform. Syst.* **18**(1), 3–13 (2021). <https://doi.org/10.2514/1.I010848>
8. Costello, D.H., Jewell, J., Xu, H.: Autonomous flight-test data in support of safety of flight certification. *J. Air Transp.* **29**(2), 93–106 (2021). <https://doi.org/10.2514/1.D0220>
9. Costello, D.: Certifying an autonomous system to complete tasks currently reserved for qualified pilots. PhD thesis, University of Maryland, College Park (2020)
10. NAVAIRINST 13034.1G, Airworthiness Policy (2022)
11. Reason, J.: Human Error. "Cambridge University press" (1990)
12. Swiss Cheese Model - Aviation Safety (2020). <https://www.aviationfile.com/swiss-cheese-model/>
13. Seshia, S.S., Bryan Young, G., Makhinson, M., Smith, P.A., Stobart, K., Croskerry, P.: Gating the holes in the swiss cheese (part i): Expanding professor reason's model for patient safety. *J. Eval. Clin. Pract.* **24**(1), 187–197 (2018)
14. Larouze, J., Le Coze, J.-C.: Good and bad reasons: The swiss cheese model and its critics. *Saf. Sci.* **126**, 104660 (2020)
15. Reason, J.T.: The Human Contribution: Unsafe Acts, Accidents and Heroic Recoveries. CRC Press, Boca Raton (2008)
16. Baird, A.R., Archibald, E.J., Marr, K.C., Ezekoye, O.A.: Explosion hazards from lithium-ion battery vent gas. *Journal of power sources.* **446**(C), 227257 (2020). Sandia National Lab. (SNL-NM), Albuquerque, NM (United States)
17. Park, Y.J., Kim, M.K., Kim, H.S., Lee, B.M.: Risk assessment of lithium-ion battery explosion: chemical leakages. *Journal of toxicology and environmental health. Part B, Critical reviews.* **21**(6–8), 370–381 (2018)
18. Zalosh, R., Gandhi, P., Barowy, A.: Lithium-ion energy storage battery explosion incidents. *Journal of loss prevention in the process industries.* **72**, 104560 (2021)
19. Kim, S.W., Park, S.G., Lee, E.J.: Assessment of the explosion risk during lithium-ion battery fires. *Journal of loss prevention in the process industries.* **80**, 104851 (2022)
20. Barone, T.L., Dubaniewicz, T.H., Friend, S.A., Zlochower, I.A., Bugarski, A.D., Rayyan, N.S.: Lithium-ion battery explosion aerosols: Morphology and elemental composition. *Aerosol Sci. Technol.* **55**(10), 1183–1201 (2021)

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